

Task 1 — Types of Malware

Malware	Main difference	Example of potential harm
Virus	Attaches itself to another file/program and usually needs the user to execute it.	A malicious attachment downloaded through an email or messaging app could infect files on a user's computer.
Worm	A self-replicating malware that can spread without attaching to another file.	A malicious attachment downloaded through an email or messaging app could infect files on a user's computer.
Ransomware	Encrypts or locks files/systems and demands payment.	Ransomware could encrypt restaurant menus, customer databases, and order records.
Trojan	Pretends to be legitimate software but performs malicious actions.	A fake food-delivery APK could steal login credentials or other information.
Spyware	Secretly monitors or collects information about the victim.	Spyware in a mobile app could collect browsing activity or sensitive information.
Adware	Displays unwanted advertisements and may track browsing activity.	Adware could display intrusive advertisements while a user is browsing a food-delivery website.

Task 2 — EICAR Test File + VirusTotal

Step 1 — Download EICAR

- Use the official EICAR download page:
- EICAR Anti-Malware Test File
- Download eicar.com or eicar.com.txt.
- Your antivirus may immediately detect/quarantine the file. That's expected.

Step 2 — Upload to VirusTotal

Open:

- VirusTotal
- Then:
- Select Upload file.
- Choose the EICAR file.
- Wait for the analysis.
- Open the Detection tab.
- Record the detection results.
- Take a screenshot.

Task 3 — Android APK VirusTotal Analysis

Procedure

- Download the APK to your computer.
- Do not install it on your personal phone.
- Go to VirusTotal.
- Upload the APK.
- Wait for the analysis.
- Open Detection.
- Open Details.
- Open Behavior if available.
- Record suspicious or noteworthy indicators.

Task 4 — Ransomware Demo in a Sandbox

Step-by-step summary

A typical ransomware sandbox demonstration works approximately as follows:

1. Initial execution
 - The ransomware sample is executed inside the isolated virtual machine.
2. Malware initialization
 - The program performs its initial malicious actions and may create files, processes, or registry/system changes.
3. File discovery
 - It searches accessible directories for files that it can potentially encrypt.
4. File encryption
 - Target files are encrypted or otherwise made inaccessible.
5. File renaming/extension changes
 - Encrypted files may receive a new extension or filename.
6. Ransom note
 - The malware creates a message explaining that the files have been encrypted and demanding payment.
7. System changes
 - Depending on the ransomware, additional changes may occur, such as modifying settings or creating persistence mechanisms.
8. Sandbox observation
 - The analyst observes the processes, file modifications, network activity, and system changes without exposing the host computer to the malware.

Task 5 — Zomato-Style Food Delivery App

1. Spyware
 - Potential impact:
 - Spyware hidden inside a malicious application could collect users' sensitive information, browsing activity, credentials, or other private data.
 - Security feature:
 - Implement strong application permissions and least-privilege access. The app should only request permissions that are genuinely required.

2. Adware

- Potential impact:
 - Adware could display unwanted advertisements, redirect users to malicious websites, consume mobile data/battery, or degrade the user experience.
- Security feature:
 - Use application integrity/security monitoring and only distribute trusted, digitally signed application builds. Perform regular security testing of third-party SDKs and advertising components.